

PVTRACE · REQUIREMENTS

PVTrace requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/pvtrace		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements with concept status for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

PVTrace requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet agreed with users, and will be checked by calculation at TRL 3 and revised after co-design (see PVT-PRB-001). The status column gives the concept's position against each target from the estimates in PVT-PRC-001. Three requirements are **not met** on current estimates (R3, R13 in part, R16) and several are unverified.

The **reference module** is a 144 half-cell module of about 450 W: open-circuit voltage (Voc) about 49.5 V, short-circuit current (Isc) about 11.6 A. The **high-current case** is a 108 half-cell 182 mm module of about 410 W (Voc about 37.5 V, Isc about 13.9 A). The **high-voltage case** is two older 60-cell 250 W modules in series (Voc about 75 V, Isc about 8.9 A).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Voltage range	0 to 100 V DC open circuit, including cold-weather Voc rise; hard limit below 120 V	Component ratings; calculation	Met by design (160 V capacitors, 150 V MOSFETs)
R2	Current range	0 to 20 A short circuit	Shunt and switch ratings	Met by design
R3	Sweep duration, to limit capacitive error in high-efficiency cells	20 to 200 ms from Isc to Voc for any module in range	Sweep model; later bench sweep	Not met: about 13 ms for the high-current case; 21 ms reference; 43 ms high-voltage case (estimates)
R4	Curve resolution	200 or more V-I pairs per sweep	Sample-rate calculation	Met (estimate): about 325 to 1,070 pairs
R5	Measurement accuracy after calibration	Voltage and current within ± 1 % of reading ± 0.1 % of full scale	Error budget; later check against a calibrated meter	Unverified
R6	Repeatability	Pmax within ± 1 % over 3 consecutive sweeps at stable irradiance (± 1 % change)	Later field test	Unverified
R7	Irradiance and temperature	Plane-of-array irradiance within ± 5 % from a reference cell; module back temperature within ± 2 °C	Calibration plan; CalRig for the temperature probe	Unverified; depends on reference-cell calibration
R8	STC translation	Pmax translated to STC per IEC 60891 with an expanded uncertainty of ± 5 % or better at 700 W/m ² or more	Uncertainty budget	At risk: estimate about ± 5 to 6 % with a once-calibrated reference cell
R9	Fault flags	Flag steps from bypass diode conduction (loss of one substring or more), raised series resistance, lowered shunt resistance and an Isc deficit against irradiance; shown within 5 s of the sweep	Rule definitions; later tests on modules with known faults	Unverified; rules drafted in PVT-PRC-001
R10	Second-life grading	Grade A, B, C or reject from STC Pmax against nameplate, with thresholds that a partner can edit	Rule review with partner	Proposed thresholds awaiting Amish
R11	Safe connection and disconnection	Reverse polarity and over-voltage detected before the load connects; the circuit is never opened above 0.5 A; load capacitors below 30 V within 2 s after each sweep and below 60 V within 60 s if the controller fails	Circuit review; discharge calculation	Met by design (estimates: about 0.1 s active to below 30 V; about 23 s passive to below 60 V)
R12	Handheld	Mass 1.5 kg or less including leads and sensor pod; enclosure 250 x 150 x 100 mm or less	Mass estimate; model	Met (estimate): about 1.35 kg, 200 x 120 x 75 mm
R13	Field conditions	Operate at 0 to 45 °C ambient in full sun; IP54 when closed; display readable in direct sun	Datasheets; later field check	IP54 and temperature met by design; sunlight readability not met with a

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
				commodity TFT (unverified)
R14	Battery life	8 h of field use or 200 sweeps per charge; USB-C charging	Power budget	Met (estimate): about 16 h
R15	Open data	Every sweep saved as CSV (raw V-I pairs, irradiance, temperature, time, module ID, flags, grade); export by USB or Wi-Fi to a phone	Firmware sketch review at TRL 3	Met by design
R16	Second-life safety screening	Detect insulation faults (cracked backsheet, wet leakage) before a module is resold	Not in this instrument	Not met: out of scope; an insulation tester is needed alongside PVTrace
R17	Cost	\$150 or less in parts for one prototype	Priced BOM (bom/ bom.csv)	Met with about \$1 margin (about \$149); galvanic isolation option would exceed it

■ Assumptions

- Module curves are modeled with a single-diode model fitted to datasheet Voc, Isc, Vmp and Imp; sweep times are the time to reach 99 % of Voc with a 4.4 mF load.
- The ADC delivers at least 25,000 V-I pairs per second in practice (half the rated rate of a 100 kS/s two-channel converter, to allow for SPI and firmware overhead).
- The reference cell is calibrated once against a pyranometer or a module of known power under clear sky; its uncertainty (about ± 3 to 5 %) dominates R8.
- The 20 ms lower bound in R3 is a working figure for high-efficiency (PERC, TOPCon, HJT) cells, whose capacitance distorts fast sweeps; the bound will be checked against literature at TRL 3.
- 700 W/m² follows IEC 61829 guidance for converting to STC ([Seaward](#)).

Safety: The tracer connects directly to energized PV modules at up to 100 V DC and 20 A, and stores up to 22 J in its load capacitors. R11 exists to keep the user safe; it must not be relaxed to meet cost.